

Monday warm Up: Unit 3, Forces II and III

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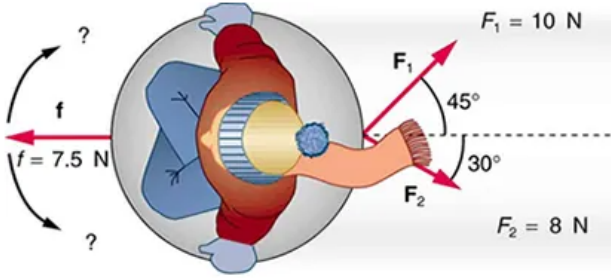


Figure 1: A child is pulled on a sled by two other children, and there is also a force of friction.

3 Forces III

- Suppose the children let go of the sled in the previous exercise, and the sled crosses onto a large patch of ice. (a) If the total mass of the rider plus the sled is 50 kg, what is the *weight* of the system? What is the normal force? (b) The force of friction is $\vec{f} = -\mu N \hat{i}$, where μ is the coefficient of friction and N is the magnitude of the normal force. If μ for sled and ice is 0.02, what is the force of friction? (c) When will the sled stop, if the net force is solely friction?

1 Memory Bank

- $\vec{s} = -k\Delta\vec{x}$... Spring force
- $\vec{f} = -\mu N \hat{i}$... Force of friction on a flat surface
- $F_D = \frac{1}{2}C\rho A v^2$... Drag force for a velocity v

2 Forces II

- A 5.00×10^5 kg rocket is accelerating straight up. Its engines produce 1.250×10^7 N of thrust, and air resistance is 4.50×10^6 N. What is the rocket's acceleration?
- Two children pull a third child on a snow saucer sled exerting forces $\vec{F}_1$ and $\vec{F}_2$ as shown from above in Fig. 1. Find the acceleration of the 50.00-kg sled and child system. Note that the direction of the frictional force will be in the opposite direction of $\vec{F}_1 + \vec{F}_2$.

- Sometimes, friction can be *static*. That is, there's enough friction to keep an object stuck in position. Suppose a sled is on a hill with incline angle θ , and the coefficient of static friction is μ_s . Show that the maximum θ , such that the sled does not move, satisfies

$$\mu_k = \tan \theta \quad (1)$$

- The force of *drag* (Memory Bank) on a diving falcon sets its maximum speed. Show that, if a falcon is diving straight down to catch prey, that the fastest speed at which it can travel is

$$v = \sqrt{\frac{2mg}{C\rho A}} \quad (2)$$

where A is the area of the falcon, ρ is the density of air, m is the mass of the falcon, and $g = 9.81 \text{ m s}^{-2}$. Let the constant $C = 1$. Estimate the maximum speed of a falcon.